

Achievable Voltage Support of Multi-Hub Vehicle-to-Grid Operation Under Fleet Energy Constraints

Md. Rifat Rahman

*Department of Electrical & Electronic Engineering
Bangladesh University of Engineering & Technology
Dhaka, Bangladesh
2006137@eee.buet.ac.bd*

Abstract—Vehicle-to-grid (V2G) operation is frequently proposed as a voltage support resource on weak distribution feeders, and recent work compares learned controllers with local droop control for this task. These comparisons are usually made without first establishing how much voltage support the connected fleet is able to supply. This paper computes a per-hour achievable ceiling for a five-hub configuration on the IEEE 34-bus feeder by optimizing per-hub active and reactive injection directly on the AC power flow, subject only to inverter rating. The ceiling is then used to separate the shortfall caused by limited stored energy from the shortfall caused by the controller. Coordinated uneven dispatch across five hubs clears all eighteen violating hours at a mild peak, compared with seven hours for a single hub of the same rating. When realistic availability and state-of-charge limits are applied, the fleet supplies only 14.5 percent of the energy requested by the unconstrained controller, and the day-integrated violation rises from 4.15 to 42.77 per-unit-hours. By comparison, the difference between a tuned droop law and a soft actor-critic policy trained for 100,000 steps is 0.79 per-unit-hours. Energy adequacy therefore outweighs controller choice by a factor of 45 to 50 at both load levels, which indicates that fleet sizing and availability deserve more attention than controller sophistication on feeders of this type.

Index Terms—vehicle-to-grid, voltage regulation, distribution networks, reinforcement learning, battery degradation, energy adequacy

I. INTRODUCTION

Rural and semi-rural distribution feeders were not designed for the voltage profiles they are now required to hold. Long radial circuits with light conductors experience considerable voltage drop under peak demand, and the conventional remedies, such as reconductoring, additional voltage regulators and capacitor banks, are slow to deploy and capital intensive. The bidirectional inverters installed with electric vehicle (EV) fleets offer an alternative. They are located towards the load end of the feeder where support is most effective, and their cost is borne by the fleet owner rather than the utility. Global EV stock continues to grow rapidly [1], and with it the notional capacity available for grid services.

Much of the recent literature has accepted this premise and moved on to the control problem. Deep reinforcement learning has been applied to inverter dispatch for voltage regulation using safe off-policy methods [2], multi-agent consensus

formulations [3], and graph-based architectures with explicit safety layers [4]. Work specific to V2G extends these ideas to coordinated multi-hub operation with battery-aware constraints [5]. The assumption common to this body of work is that the controller is the binding constraint, so that an improved policy will extract more voltage support from the same hardware.

This assumption has not been examined on the systems where it is most often applied. None of the studies cited above computes what the fleet could achieve under an ideal controller, so no reported result can be placed on a scale. A learned policy that halves the violation of a droop baseline may have captured most of the available benefit or only a small part of it, and the published numbers do not distinguish between the two cases.

This paper supplies the missing reference. For each hour of a daily load profile, the per-hub active and reactive injection that minimizes a two-sided voltage violation metric is obtained directly from the AC power flow, subject only to inverter rating. The result is an energy-unconstrained ceiling that describes what the installed power electronics could deliver if the batteries behind them were not limiting. Comparing this ceiling with droop and learned controllers operating with and without a realistic fleet model separates the total shortfall into a component due to stored energy and a component due to control.

The main contributions of this paper are as follows:

- A per-hour achievable ceiling for multi-hub V2G voltage support on the IEEE 34-bus feeder, obtained by coordinate descent on the AC power flow, and the separation of energy and control contributions that it permits (Section III).
- A measurement of the active and reactive trade-off on a stored-energy basis rather than an apparent-power basis, which is the appropriate basis when state of charge is drawn down by apparent power (Section III-C).
- A controller comparison under realistic availability and state-of-charge limits, including a training-budget sweep that distinguishes genuine controller limitations from insufficient training (Section IV).

TABLE I
IDENTIFYING THE REGULATOR SETTING FROM THE NO-V2G BASELINE

Setting	Mild		Aggressive	
	Mean	Min	Mean	Min
Reported [5]	0.963	0.907	0.883	0.807
Regulators off (this work)	0.951	0.903	0.860	0.800
Regulators active (this work)	1.012	0.983	0.911	0.831

- A degradation-aware dispatch frontier, together with three negative results concerning safety projection, state augmentation and droop fixed-point convergence (Sections V and VI).

The remainder of the paper is organized as follows. Section II describes the feeder, the fleet and battery model, the evaluation metrics and the controllers, and reproduces the baseline of the reference study. Section III computes the achievable ceiling and separates the energy and control contributions. Section IV reports controller performance under fleet constraints. Section V presents the degradation-aware frontier, Section VI reports three negative results, and Section VII concludes.

II. SYSTEM MODEL AND STUDY DESIGN

A. Feeder and Charging Hubs

All experiments use the IEEE 34-bus test feeder [6], a 24.9 kV rural circuit whose length makes it a standard stress case for voltage support studies. The feeder is solved in OpenDSS [7] through its direct Python interface. The single-hub configuration places one 500 kW/400 kVAr hub at bus 890. The multi-hub configuration distributes five hubs of the same rating at buses 890, 844, 832, 830 and 860, following the reference study [5].

A daily load shape drives hours 06:00 through 23:00, scaled by a peak multiplier. Two operating points are reported throughout: a mild peak ($\lambda_{\max} = 1.5$) and an aggressive peak ($\lambda_{\max} = 3.0$), also following [5].

B. Reproduction of the Reference Baseline

The reference study does not state whether the voltage regulators of the IEEE 34-bus feeder are active, and the answer changes every number that follows. Its reported no-V2G statistics identify the setting. Those statistics are computed from hourly mean bus voltages, so that convention is matched here only; elsewhere in this paper voltages are evaluated per phase, for the reason given in Section II-D.

Table I settles the question. With the regulators disabled the minimum voltages fall within 0.004 to 0.007 p.u. of the reported values, whereas with them active the mild minimum differs by 0.076 p.u., which is outside any plausible modelling difference. The feeder mean obtained here runs 0.012 to 0.023 p.u. below the reported mean, which is attributed to load allocation details not specified in the reference, so the configuration is identified from the minimum rather than from the mean. All subsequent results use disabled regulators.

C. Fleet and Battery Model

Each hub serves a fleet whose availability varies through the day between 45 and 85 percent and is drawn independently per scenario. Vehicles arrive at 70 percent state of charge and are not discharged below 20 percent. State of charge is drawn down by apparent power, so reactive support costs the same stored energy as active support. This follows the reference formulation and is significant for Section III-C.

The distinction between an unconstrained and a constrained fleet runs through the whole study. The reference work evaluates multi-hub coordination while “assuming sufficient EV capacity at each hub, allowing the analysis to isolate the benefits of multi-hub coordination from availability constraints” [5]. Both cases are reported here, because the difference between them turns out to be the principal result.

D. Evaluation Metrics

ANSI C84.1 [8] is a per-phase standard, so voltages are evaluated per phase rather than as bus averages, which mask single-phase excursions on this unbalanced feeder. The primary metric is the day-integrated two-sided violation

$$IV = \sum_h \sum_p \left[\max(0, v_{\min} - v_{p,h}) + \max(0, v_{p,h} - v_{\max}) \right] \Delta t \quad (1)$$

in per-unit-hours, with $v_{\min} = 0.95$, $v_{\max} = 1.05$ and $\Delta t = 1$ h. Unlike a count of violation hours, (1) does not saturate when two controllers violate in the same hours by different margins, and being two-sided it penalizes overvoltage as well as undervoltage.

Every comparison uses common random numbers, so that all controllers see an identical availability draw, initial state of charge and load realization, and differences are paired. Confidence intervals are 95 percent over the paired scenarios and training seeds.

E. Controllers

The droop baseline is a Volt-Watt and Volt-VAr characteristic with a ± 0.02 p.u. deadband, consistent with IEEE 1547 [9]. Because droop response and feeder voltage are mutually dependent, the baseline is not evaluated open-loop. The fixed point is obtained by damped Jacobi updates with a residual convergence test, a detail examined further in Section VI.

The learned controller is soft actor-critic [10] as implemented in Stable-Baselines3 [11], with two hidden layers of 256 units, a learning rate of 3×10^{-4} and a batch size of 256. Episodes span one day, so that intertemporal rationing of a finite battery is representable. The state comprises bus voltages, load multiplier, hour of day, and per-hub state of charge and availability. The action sets per-hub active and reactive power as a residual on the droop prior.

III. ACHIEVABLE VOLTAGE SUPPORT

A. Achievable Ceiling

For each hour, the per-hub injection (P_i, Q_i) that minimizes (1) subject to $\sqrt{P_i^2 + Q_i^2} \leq S_i^{\text{rated}}$ is obtained by coordinate

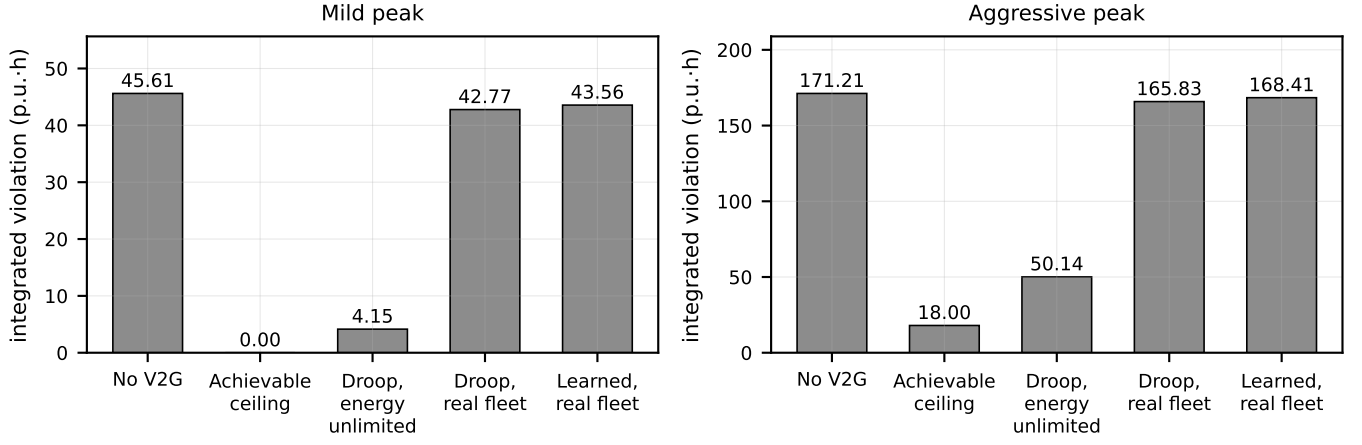


Fig. 1. Day-integrated two-sided voltage violation on the five-hub feeder. The achievable ceiling is the optimized per-hub dispatch bounded only by inverter rating. The distance between the energy-unlimited and real-fleet droop bars is the cost of limited fleet energy; the distance between the two real-fleet bars is the full span of controller choice.

TABLE II
ACHIEVABLE CEILING UNDER OPTIMIZED PER-HUB DISPATCH

Configuration	Clearable hours (of 18)	Day-total IV (p.u.·h)	
		Uniform	Optimized
Single hub, mild	7	11.14	10.78
Single hub, aggressive	2	125.84	123.97
Five hubs, mild	18	0.29	0.00
Five hubs, aggressive	4	28.33	18.00

descent over a polar grid with refinement, evaluating a full AC power flow at every candidate. No energy constraint is imposed. Non-converged solutions are masked rather than read, because at deep-sag hours the high-injection end of the search lies past the nose of the power-voltage curve, where OpenDSS leaves stale values in its arrays.

Searching per hub rather than uniformly is necessary. With five hubs at equal output, raising the feeder end to 0.95 p.u. drives the near buses above 1.05 p.u., so hours that an uneven dispatch clears comfortably are scored as unclearable. Under uniform dispatch the five-hub case then scores worse than the single-hub case, which is not physically possible. Per-hub search removes the artefact and moves the mild multi-hub result from five clearable hours to eighteen.

Table II gives the result. A single hub at full rating clears seven of the eighteen loaded hours at the mild peak and two at the aggressive peak. Five hubs of the same individual rating, dispatched unevenly, clear all eighteen and four respectively. Fig. 2 shows the hourly picture. The reachable envelope sits on or above the 0.95 p.u. limit for the entire mild day, and falls short of it for eleven hours of the aggressive day regardless of how the inverters are dispatched. Those eleven hours are a siting and rating problem rather than a control problem, and no policy can clear them.

The value of coordination is therefore quantifiable rather than assumed. At equal total inverter rating, distributing ca-

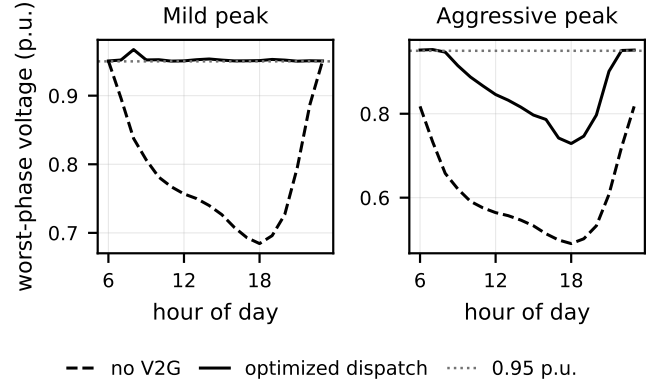


Fig. 2. Worst-phase voltage by hour with no V2G and under optimized per-hub dispatch. The mild-peak envelope clears the limit for the whole day; eleven aggressive-peak hours cannot be cleared at any dispatch.

capacity across five hubs and dispatching it unevenly reduces day-integrated violation by 100 percent at the mild peak and by 36.5 percent at the aggressive peak, relative to uniform dispatch.

B. Separation of Energy and Control Contributions

Fig. 1 places the controllers against this ceiling and is the central result of the paper. Taking the mild panel, the untreated feeder accumulates 45.61 p.u.·h of violation and the ceiling removes all of it. A droop law with unlimited stored energy removes 41.46, or 91 percent, while the same droop law with a realistic fleet removes 2.84, or 6 percent. The fleet delivers 1342 kWh against the 9264 kWh requested by the unconstrained law, a ratio of 14.5 percent. At the aggressive peak the corresponding ratio is 5.7 percent.

The controller span lies inside that shortfall. Between tuned droop and the best learned policy obtained here, the difference is 0.79 p.u.·h at the mild peak and 2.58 at the aggressive peak,

TABLE III
FIVE-HUB FEEDER WITH REALISTIC FLEET CONSTRAINTS

Controller	Mild		Aggressive	
	IV (p.u.·h)	Throughput (kWh)	IV (p.u.·h)	Throughput (kWh)
No V2G	45.61	0	171.21	0
Droop, real fleet	42.77	1342	165.83	1582
Learned, 20k steps	45.45	5842	169.60	3227
Learned, 100k steps	43.56	5588	168.41	3176
Droop, unlimited	4.15	9264	50.14	27814

against energy shortfalls of 38.62 and 115.69 respectively. Energy adequacy outweighs controller choice by a factor of 49 and 45.

C. Active and Reactive Power per Unit of Stored Energy

Because state of charge is drawn down by apparent power, the usual assumption that reactive power is free does not apply. Sweeping the injection angle at fixed apparent power and fixed loading shows that active power delivers between 1.25 and 1.49 times the worst-phase voltage rise of reactive power for the same battery energy withdrawn. The voltage-optimal split is 35 to 40 degrees from the active axis at half rating, and migrates towards 17 to 26 degrees at full rating as the resistive voltage drop becomes dominant. For reference, the hub rating ratio of 400 kVAr to 500 kW corresponds to 38.7 degrees.

This has a direct design consequence. On a feeder with this R/X character, a V2G hub sized by apparent power and operated at its rating ratio commits more reactive power than is useful, and pays full battery energy for it.

IV. CONTROLLER PERFORMANCE UNDER FLEET CONSTRAINTS

A. Multi-Hub Operation

Table III reports the five-hub feeder with the fleet model active, over three seeds and five paired scenarios. Droop reaches 42.77 p.u.·h at the mild peak and 165.83 at the aggressive peak. The learned policy trained on the conventional 20,000-step budget reaches 45.45 and 169.60, losing to droop in every paired scenario at both load levels.

Two features of the table are worth noting. First, the learned policy consumes 5842 kWh of battery throughput at the mild peak against 1342 kWh for droop, and converts that into a worse voltage outcome; at the mild peak its result is statistically indistinguishable from installing no V2G at all. Second, both controllers respect the upper voltage limit exactly, at 1.050 p.u., a point returned to in Section VI.

B. Effect of Training Budget

A negative result about a learned controller invites the objection that it was not trained for long enough, and the objection is a reasonable one, since 20,000 steps is approximately 1100 episodes and 19,000 gradient updates for a ten-dimensional action space. The policy was therefore trained to 100,000 steps and evaluated at 20,000-step checkpoints

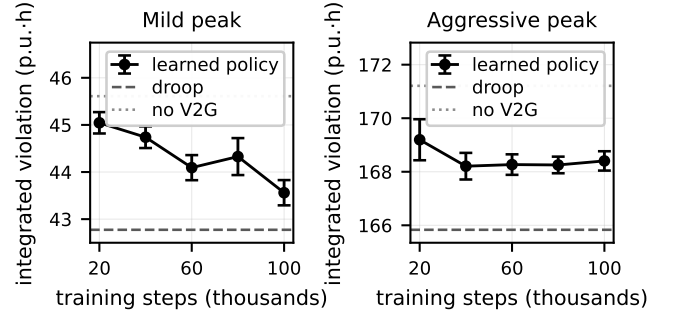


Fig. 3. Violation against training budget, with five paired scenarios and two seeds per point. The aggressive-peak curve plateaus by 40,000 steps; the mild-peak curve is still improving at 100,000.

on identical scenarios. Checkpoints were placed on the same chunk boundaries used by the fixed-budget runs, so that the 20,000-step row measures the same agent reported in Table III.

The outcome differs by load level, as Fig. 3 shows. At the aggressive peak the curve is flat from 40,000 steps onward. Violation moves from 169.20 to 168.41 across a fivefold increase in budget, which lies inside the combined confidence interval of 1.13, and droop wins every paired scenario at every checkpoint. The advantage of droop under aggressive stress is therefore not an artefact of training budget.

At the mild peak the curve is still improving at 100,000 steps. Violation falls from 45.05 to 43.56, a change of -1.48 against a confidence interval of 0.49, the gap to droop narrows monotonically from $+2.27$ to $+0.79$, and the policy takes two of ten paired scenarios, which are the only scenarios it wins anywhere in this study. The mild comparison is therefore reported as unresolved at the budgets tested rather than as a droop victory.

It is worth placing the improvement on the scale of the previous section. The gap narrowed by 1.48 p.u.·h over a fivefold increase in training, against an energy shortfall of 38.62 at the same operating point. Additional training moves the policy within a band that remains small compared with the constraint governing the outcome.

C. Active and Reactive Allocation

A scoreboard alone does not explain the outcome, so the active and reactive split commanded by the policy was examined hour by hour, weighted by apparent power and compared against the voltage-optimal angle recomputed at the injection level the policy itself selected.

The policy does not track the optimum, and the deviation does not fall with additional training. At 20,000 steps the mean absolute deviation is 26.5 degrees at the mild peak and 32.5 degrees at the aggressive peak. At 100,000 steps these become 20.8 and 31.5 degrees. The changes of -5.7 and -1.0 degrees both lie inside their combined 95 percent confidence intervals of 24.2 and 35.1 degrees, so neither is a reduction the data supports.

Fig. 4 shows the hourly picture at 100,000 steps. The voltage-optimal angle stays between 29 and 35 degrees for most of the day at both load levels. At the mild peak the commanded angle stays near 20 degrees, with an apparent-power-weighted daily mean of 18.8 degrees. At the aggressive peak the daily mean falls to 1.2 degrees, which corresponds to almost pure active power, and in 50 percent of sampled hub-hours the commanded reactive power is negative. Reactive absorption lowers voltage on this feeder, so these are actions of the wrong sign for the condition being corrected rather than merely conservative ones. The pattern as a whole is consistent with a controller that scales a roughly fixed power factor rather than allocating between active and reactive power in response to loading.

V. DEGRADATION-AWARE DISPATCH

Battery wear is the standing objection to V2G participation, and the economic case is sensitive to it [12]. The reference formulation carries no battery term in its reward. An amp-hour throughput cost was added, expressed on the same scale as the voltage penalty, and its weight swept across three orders of magnitude.

The resulting frontier is flat, as Fig. 5 shows. Across a 300-fold sweep the day-integrated violation stays within a 2.6 percent band, between 44.83 and 46.01 per-unit-hours, while daily battery throughput falls from 6188 to 2449 kWh. Comparing the endpoints, roughly 60 percent of battery throughput can be removed for a 1.1 percent increase in violation.

This follows from the energy-limited regime identified earlier. When the binding constraint is that the fleet cannot supply what the feeder needs, the marginal voltage value of the last kilowatt-hour discharged is very low, so a wear penalty removes it at almost no cost. The practical reading is favourable: on feeders in this regime, aggressive degradation-aware dispatch costs very little in voltage performance, which improves the participation case for fleet owners.

The intermediate points should not be over-interpreted. Each weight was trained with a single seed and the sequence is not monotone, so the defensible claims are the endpoint comparison and the width of the band rather than the shape in between.

VI. ADDITIONAL OBSERVATIONS

This section reports three checks that did not produce the expected effect. They are included because each addresses a design choice that is common in the literature.

A. Safety Projection

Projecting commanded setpoints onto the voltage-feasible set is standard practice in learning-based volt/var control [2], [4], so its absence would be a reasonable objection. Two projections were implemented and measured: a proportional reduction along the commanded ray, and a reactive-first variant motivated by the result of Section III-C.

Both are inert. The unprojected policy already reports zero overvoltage and a maximum phase voltage of exactly

1.050 p.u. at both load levels, and projection changes the violation metric by less than the confidence interval. The reason is that the feasibility test acts on commanded setpoints, but the state-of-charge limit of the fleet clips those commands first. The energy constraint binds before the voltage constraint does, leaving nothing for a voltage projection to remove. Where overvoltage does appear in these runs it tracks the training regime rather than the controller: policies trained on independently sampled load multipliers without a fleet in the loop reach 1.229 p.u., whereas day-structured training with the fleet in the loop holds 1.050 across every configuration and checkpoint tested.

B. Fleet State in the Observation

Augmenting the observation with state of charge and availability, information the reference formulation does not provide, does not improve voltage performance. At the mild peak the augmented agent reaches 45.41 ± 0.36 against 44.91 ± 0.29 for a voltage-only observation, an overlapping interval. What the augmentation changes is energy use: the augmented agent spends 6068 kWh against 2497 kWh for no violation benefit, while the voltage-only agent breaches the upper limit at the aggressive peak where the augmented agent does not.

C. Droop Fixed-Point Convergence

The final observation is methodological and affects any study using a closed-loop droop baseline. Droop response and feeder voltage are mutually dependent, and the fixed point is commonly computed with a fixed iteration count. On the five-hub feeder, at a damping factor of 0.6 the iteration does not settle but enters a period-two limit cycle, with total active power alternating between 1298 and 630 kW at the mild-peak evening hour. A fixed iteration count then returns whichever branch the last iteration happened to reach, and the reported baseline flips with the parity of the iteration count. Damping of 0.3 with a residual test converges in all four configurations tested. A convergence diagnostic should therefore be reported alongside any droop baseline computed in this way.

VII. CONCLUSION

This paper computed an achievable ceiling for multi-hub V2G voltage support on the IEEE 34-bus feeder and used it to separate the shortfall due to control from the shortfall due to stored energy. Uneven coordination across five hubs is of genuine value, clearing every violating hour at a mild peak where a single hub of equal rating clears seven of eighteen. Once realistic availability and state-of-charge limits are applied, however, the fleet supplies well under a fifth of the energy the controller requests, and the resulting shortfall exceeds the entire span between a tuned droop law and a converged learned policy by a factor of roughly 45 to 50.

This is not an argument against learned control in distribution systems. It is an argument that the operating regime should be established before the controller is designed, because in the energy-limited regime an improved policy has very little left to win. Both the achievable ceiling and an energy adequacy

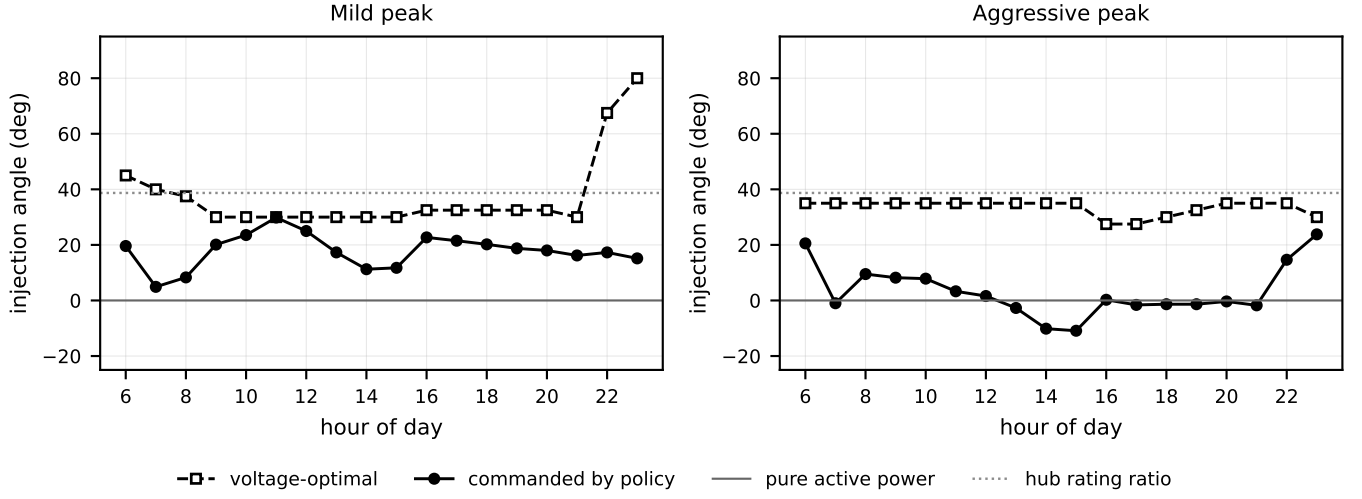


Fig. 4. Commanded injection angle against the voltage-optimal angle, by hour, at 100,000 training steps and averaged over two seeds. Zero degrees is pure active power, ninety is pure reactive, and negative values indicate reactive absorption.

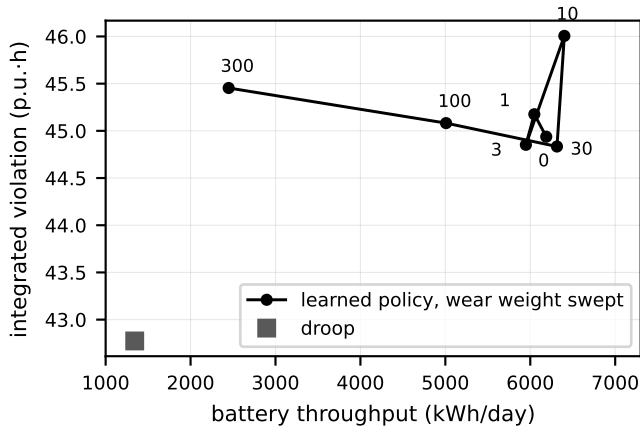


Fig. 5. Violation against battery throughput at the mild peak as the degradation weight is swept from 0 to 300 (point labels). Violation stays inside a 2.6 percent band while throughput falls by 60 percent.

ratio can be computed from feeder and fleet data before any controller exists, and it is suggested that they be reported as standard context alongside controller comparisons.

Two further results point in the same direction. A degradation-aware objective removes 60 percent of battery throughput for a 1.1 percent violation penalty, precisely because the marginal kilowatt-hour is nearly worthless in this regime, which is favourable for fleet owners weighing participation. A safety projection layer, standard elsewhere, is measurably inert here because the energy constraint binds first.

The study covers one feeder and one learning algorithm, and the mild-peak controller comparison remains unresolved at the training budgets that could be afforded. Two extensions follow directly. The ratio reported here depends on feeder impedance and hub siting, so it should be recomputed on a larger feeder.

More usefully, the ceiling computation is inexpensive enough to invert: instead of scoring a controller against it, it can serve as a planning screen that returns the fleet size and availability required to make a target violation level reachable.

REFERENCES

- [1] International Energy Agency, *Global EV Outlook 2025*. Paris, France: IEA, 2025.
- [2] W. Wang, N. Yu, Y. Gao, and J. Shi, "Safe off-policy deep reinforcement learning algorithm for volt-var control in power distribution systems," *IEEE Trans. Smart Grid*, vol. 11, no. 4, pp. 3008–3018, Jul. 2020.
- [3] Y. Gao, W. Wang, and N. Yu, "Consensus multi-agent reinforcement learning for volt-var control in power distribution networks," *IEEE Trans. Smart Grid*, vol. 12, no. 4, pp. 3594–3604, Jul. 2021.
- [4] R. Yan, Q. Xing, and Y. Xu, "Multi-agent safe graph reinforcement learning for PV inverters-based real-time decentralized volt/var control in zoned distribution networks," *IEEE Trans. Smart Grid*, vol. 15, no. 1, pp. 299–311, Jan. 2024.
- [5] J. Wang, R. A. Jacob, H. D. Kaushik, and J. Zhang, "Reinforcement learning for vehicle-to-grid voltage regulation: Single-hub to multi-hub coordination with battery-aware constraints," 2026, arXiv:2603.07237.
- [6] W. H. Kersting, "Radial distribution test feeders," *IEEE Trans. Power Syst.*, vol. 6, no. 3, pp. 975–985, Aug. 1991.
- [7] R. C. Dugan and T. E. McDermott, "An open source platform for collaborating on smart grid research," in *Proc. IEEE Power Energy Soc. Gen. Meeting*, Detroit, MI, USA, 2011, pp. 1–7.
- [8] *American National Standard for Electric Power Systems and Equipment—Voltage Ratings (60 Hz)*, ANSI C84.1-2020, 2020.
- [9] *IEEE Standard for Interconnection and Interoperability of Distributed Energy Resources with Associated Electric Power Systems Interfaces*, IEEE Std 1547-2018, 2018.
- [10] T. Haarnoja, A. Zhou, P. Abbeel, and S. Levine, "Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor," in *Proc. 35th Int. Conf. Mach. Learn.*, Stockholm, Sweden, 2018, pp. 1861–1870.
- [11] A. Raffin, A. Hill, A. Gleave, A. Kanervisto, M. Ernestus, and N. Dormann, "Stable-Baselines3: Reliable reinforcement learning implementations," *J. Mach. Learn. Res.*, vol. 22, no. 268, pp. 1–8, 2021.
- [12] M. Dubarry, A. Devie, and K. McKenzie, "Durability and reliability of electric vehicle batteries under electric utility grid operations: Bidirectional charging impact analysis," *J. Power Sources*, vol. 358, pp. 39–49, Aug. 2017.